
Casual-to-member conversion: a behavioural segmentation

Full analysis: behaviour, method, segments, and a conversion plan

Chicago bike-share · 5.84 M rides · 12 months (Jun 2025 - May 2026) · Divvy public data

■ Casual riders ■ Annual members

What this deck covers

01

The question

Why conversion, and what we ask of the data

02

How casual differs

Duration, timing and the weekday rhythm

03

Method

Features, clustering, and choosing K

04

The three segments

Profiles ranked by member similarity

05

Priority & actions

Where to focus and concrete recommendations

06

Limits & next steps

What the analysis can and cannot say

Which casual riders are worth converting?

Cyclistic has two user types — **casual riders** (single / day pass) and **annual members**. Converting the right casual riders is the cheapest growth lever.

- 01 Do casual riders behave differently from members?
- 02 Are casual riders one group — or several distinct profiles?
- 03 Which profiles look most like members, and are big enough to target?

Note Segmentation is at ride level — no persistent user ID — so clusters describe patterns of behaviour, not individual people.

12 months of Chicago bike-share trips

5.84 M

rides analysed

12

months (rolling window)

13

fields per ride

Divvy

public open data

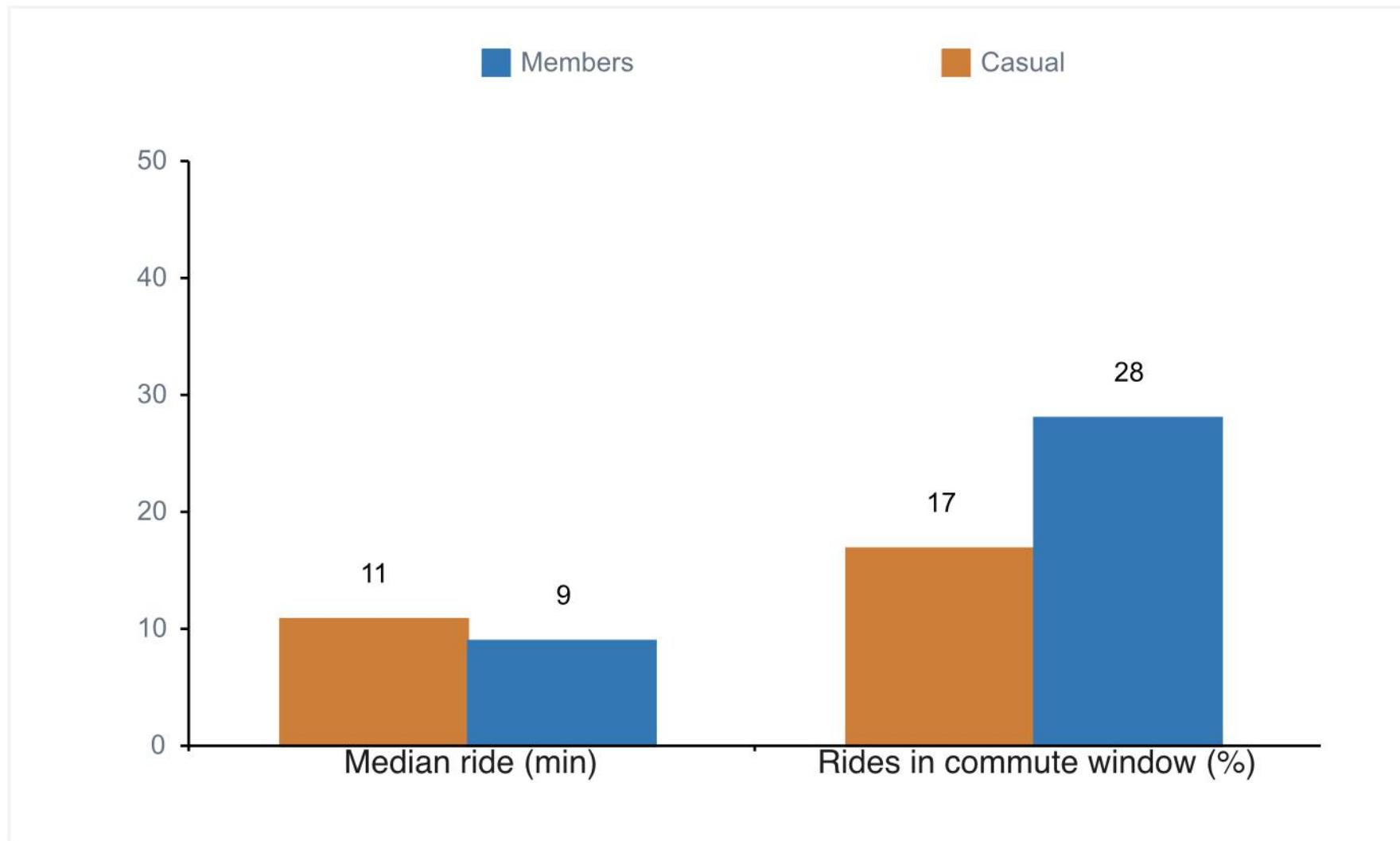
The split: of every 100 rides, **~36 are casual**.

Members 64.1%

Casual 35.9%

- Cleaned in a separate, documented pipeline (notebook 01).
- Rides with missing end-GPS and timestamp errors removed (<0.2%).
- Free-floating e-bikes legitimately have no end-station — kept by design.
- Features engineered: duration, distance, speed, weekend, commute window, e-bike, season.

Casual and member riders use the service differently



Longer casual rides

Casual median is 11 min vs 9 for members — and the casual mean is pulled far higher by long recreational trips.

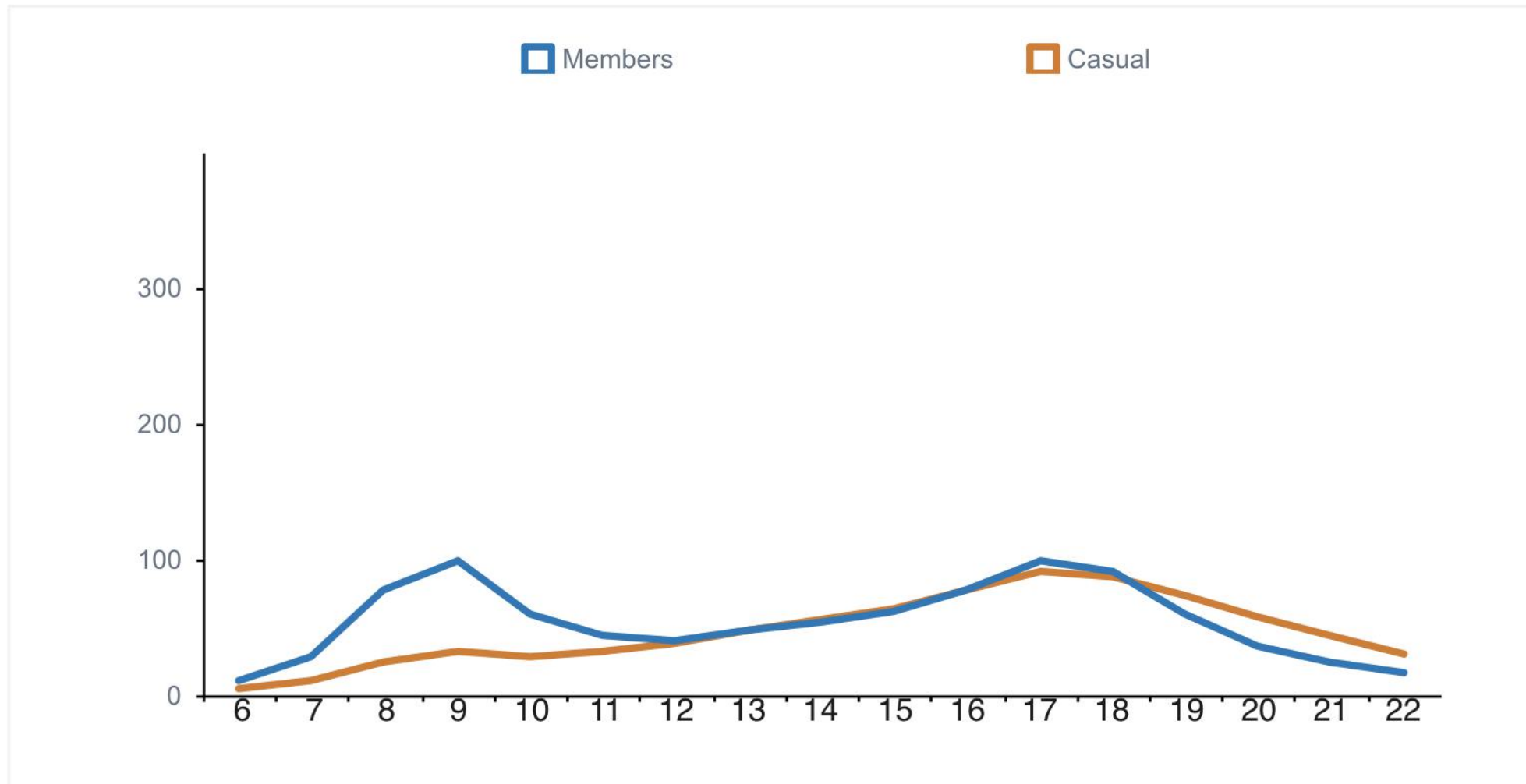
Members commute more

28% of member rides fall in commute windows vs 17% for casual.

Casual skews to weekends

Recreation, not routine, dominates casual demand.

The morning peak is where members stand alone



WHAT TO SEE

Members show two sharp peaks — morning and evening commute.

Casual riders **miss the morning peak** almost entirely, but share the evening one.

The AM weekday peak is the cleanest signal of member-like behaviour.

Hour of departure (weekdays). Schematic of the documented pattern — relative, not to scale.

How the segments were found

1

Engineer features

8 behavioural features per ride: duration, distance, speed, weekend, commute window, time-of-day, season, e-bike.

2

Cluster casual rides

MiniBatch K-Means groups casual rides into behavioural profiles — letting the data define the segments.

3

Benchmark vs members

Members are clustered the same way; each casual segment is scored by distance to the member profile.

4

Prioritise

Combined score = similarity × ride share — balancing behavioural fit against audience size.

Why three segments

K was chosen empirically — inertia and silhouette compared across $K = 2, 3, 4$. The inertia curve bends at $K=3$ (the $2 \rightarrow 3$ drop is more than twice the $3 \rightarrow 4$ drop). $K=2$ would only split weekday vs. weekend — too coarse to act on.

K = 2

TOO COARSE

Just weekday vs. weekend — not actionable.

K = 3

CHOSEN

Elbow supports it; three profiles a team can act on.

K = 4

MARGINAL GAIN

Little extra separation for added complexity.

Note Silhouette scores are modest in absolute terms — segments are soft behavioural regions, not sharply separated groups. Useful for targeting, not hard borders.

Three profiles, ranked by member similarity

E-Bike Riders

HIGH PRIORITY

52.1%

of casual rides

0.65

similarity

Short, purposeful trips on weekdays.
Already ride like members.

Commute E-Bike Riders

MEDIUM PRIORITY

17.2%

of casual rides

0.49

similarity

100% rush-hour rides; ~1 in 4 ends
outside a dock.

Classic Leisure Riders

LOW PRIORITY

30.8%

of casual rides

0.00

similarity

Longer recreational rides, weekend-
skewed.

E-Bike Riders

HIGH PRIORITY

52.1%

of all casual rides

0.65

member similarity (0–1)

The largest casual segment — and the one that already rides like a member.

- Short, purposeful trips concentrated on weekdays.
- Predominantly electric bikes, used for getting somewhere — not sightseeing.
- Closest of all casual segments to the member behavioural profile.
- Size and fit align: the obvious lead target.

ACTION

Lead the annual-membership campaign here. Trigger on short weekday e-bike rides.

Commute E-Bike Riders

MEDIUM PRIORITY

17.2%

of all casual rides

0.49

member similarity (0–1)

A rush-hour segment with a physical-infrastructure story attached.

- 100% of rides fall in commute windows; ~70% are on e-bikes.
- About 1 in 4 rides ends outside a docking station (free-floating).
- A free-floating park-up means no guaranteed bike near home next morning.
- Smaller and a step further from members than E-Bike Riders.

ACTION

Pair a membership offer with a docking pilot in residential areas where rides end.

Classic Leisure Riders

LOW PRIORITY

30.8%

of all casual rides

0.00

member similarity (0–1)

Recreational riders whose behaviour is furthest from membership.

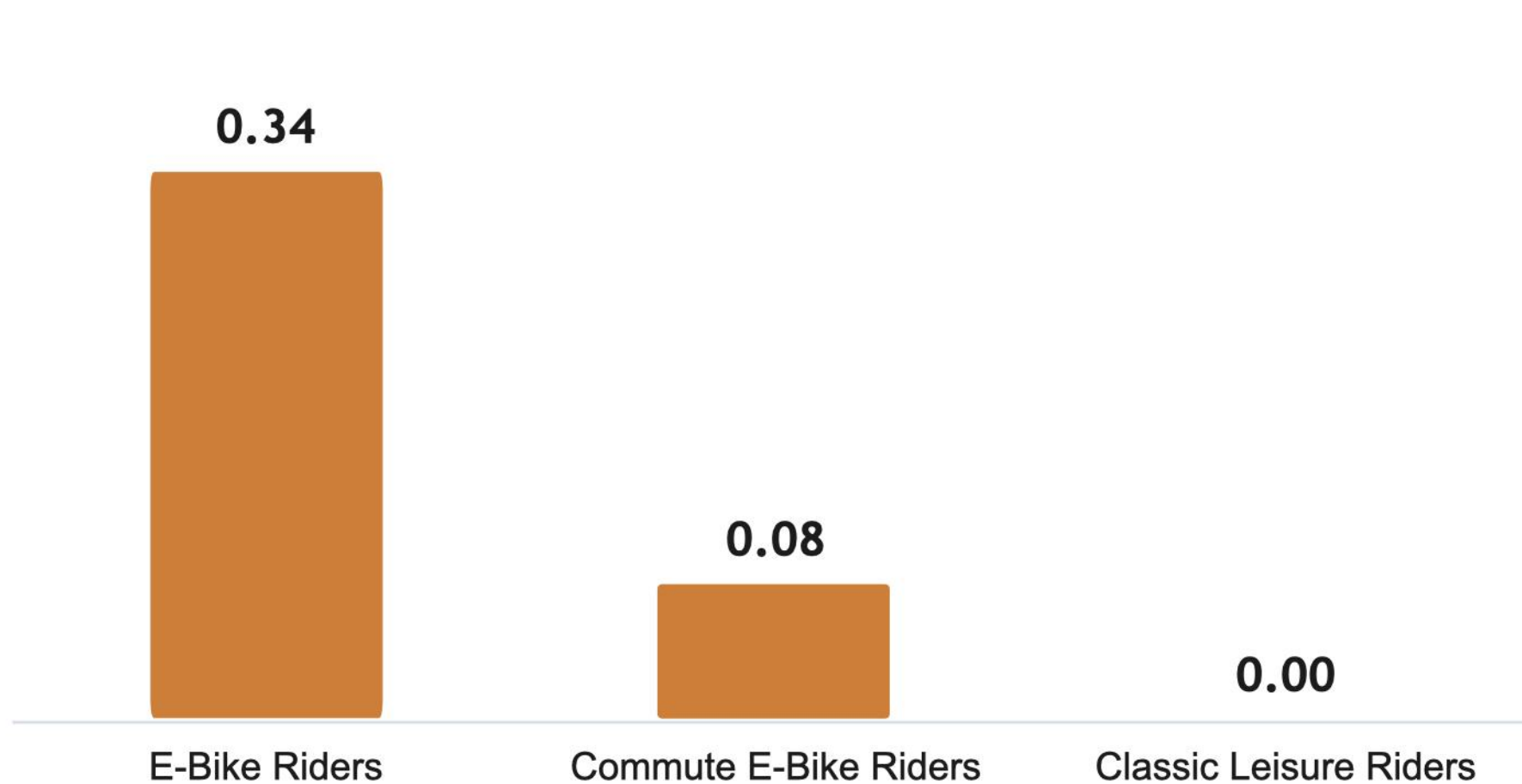
- Longer rides, classic bikes, concentrated on weekends.
- Use the service for leisure, not routine transport.
- Most distant from the member profile in this dataset (similarity 0.00).
- A sizeable group — but a poor fit for an annual-membership pitch.

ACTION

Do not push annual membership. Offer a weekend or seasonal pass instead.

Balancing fit against audience size

Priority = member similarity × share of casual rides. Both dimensions point at the same segment.



THE TAKEAWAY

E-Bike Riders score ~4× the next segment.

They are both the largest casual segment and the most member-like — so volume and fit reinforce each other rather than trade off.

Classic Leisure Riders score ~0: large, but behaviourally too far for membership.

Four moves, matched to behaviour

01

Lead with E-Bike Riders

Push annual membership to this segment first; message commute-cost savings vs. pay-per-ride.

02

Sell leisure riders a different product

Offer Classic Leisure Riders a weekend or seasonal pass — not annual membership.

03

Test the infrastructure lever

Pilot added docking where PM commute e-bike rides end; it may remove a real conversion barrier.

04

Prove it before scaling

Run a controlled A/B campaign and measure real conversion before committing budget.

The right offer for each segment

SEGMENT	OFFER	CHANNEL SIGNAL	EXPECTATION
 E-Bike Riders	Annual membership	Short weekday e-bike trips	Lead target — scale after test
 Commute E-Bike Riders	Membership + docking pilot	Rush-hour, free-floating ends	Watch — infra may unlock
 Classic Leisure Riders	Weekend / seasonal pass	Long weekend rides	Don't push annual

Test before you scale

PHASE 1

Pilot

Controlled A/B membership offer to E-Bike Riders. Hold-out group; measure real conversion, not similarity.

PHASE 2

Learn

Validate the infrastructure signal — map free-floating endpoints; pilot docking in dock-sparse residential areas.

PHASE 3

Scale

Roll out what converts. Launch a separate weekend/seasonal product for Classic Leisure Riders.

Limitations of the analysis

No user ID

Segmentation is ride-level; one person can appear in several segments. Sizes count rides, not people.

Soft clusters

Modest silhouette scores: segments are behavioural regions, not sharply separated groups.

~33% missing station data

GPS used instead; some station-based extensions aren't available.

Similarity $\neq$ conversion

It is a structural indicator. Price, alternatives and habit — not in this data — drive real decisions.

Commute is a proxy

Departure time approximates intent; commute windows also carry a high mechanical base rate (~28%).

Straight-line distance

Haversine underestimates true cycling distance, so speed is approximate.

■ TO SHARPEN THE NEXT ROUND

What would make this sharper



User IDs

Segment by person, not by ride



Survey data

Validate intent behind the behaviour



Conversion history

Calibrate similarity against real outcomes



THE DECISION

Run a controlled membership campaign on **E-Bike Riders**
— then scale on results.